Day 3
Learning representations

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Behavioral vs. representational fit

A fit between model and human behavior doesn’t tell us much about the similarity of their *representations*.

One logical consequence: if a model shows the same behavior as humans but uses different representations (compared to what humans are believed to have), then the assumed representations are not a necessary condition for successful learning.

Behavioral vs. representational fit

A theory claims humans use representational format C (e.g., discrete categories), and that C explains behavior B (e.g., a discrimination pattern).

A model reproduces B using representational format P (e.g., a continuous space), which is different from C.

Conclusion: C is not necessary for B. The behavior can arise without the assumed format.

Existence/feasibility demonstration

- **Important:** modeling evidence only presents a possibility
  - Cf. Rumelhart & McClelland: didn't establish that rules are absent, showed that rules need not be stored as such
- How to demonstrate a possible alternative for representations?
  A. Define the alternative format explicitly
  B. Let the model infer suitable representations itself

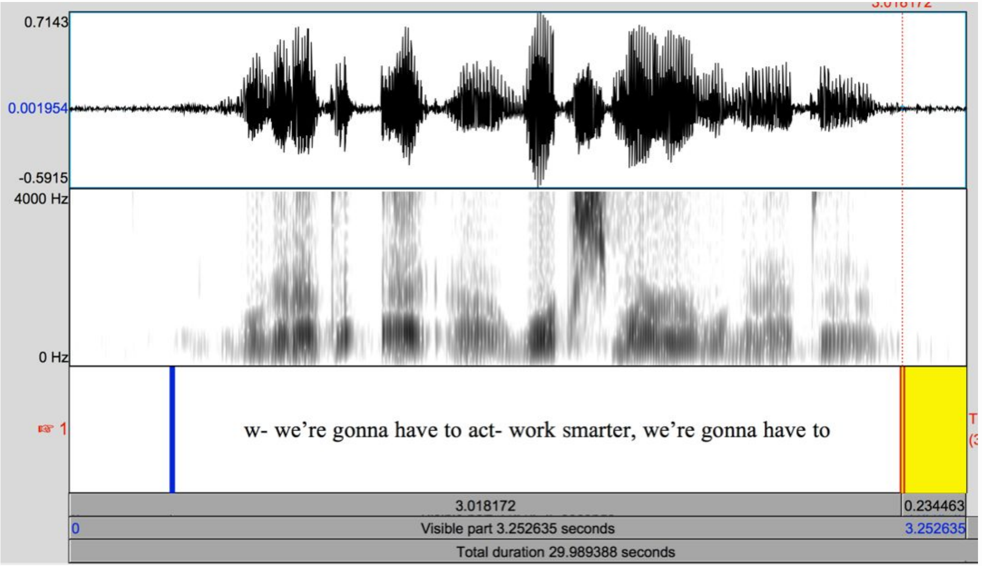
Representation learning

- Feature engineering vs. feature (or representation) learning
- In representation learning, the model works out its own way of encoding the input, instead of using predefined features
- How much is “engineered” is a continuum
  - e.g., Wickelfeatures vs. modern ASR model features
- Not only neural networks: clustering algorithms, dimensionality reduction

Categorical perception

Sound as a frequency spectrum

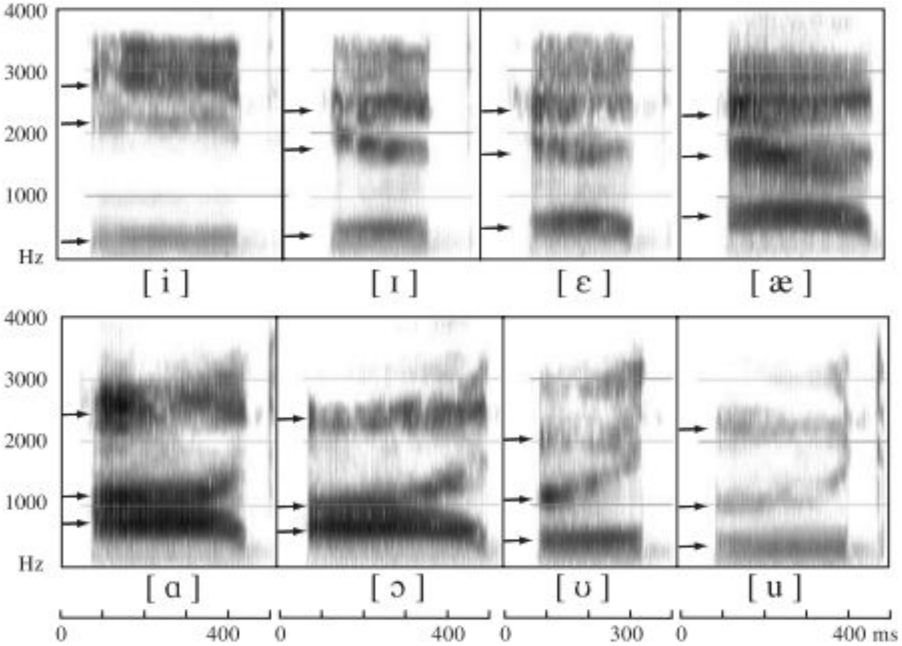
Waveforms and spectrograms



https://www.ling.upenn.edu/courses/ling0001/PraatInstructions.html

Sound as a frequency spectrum

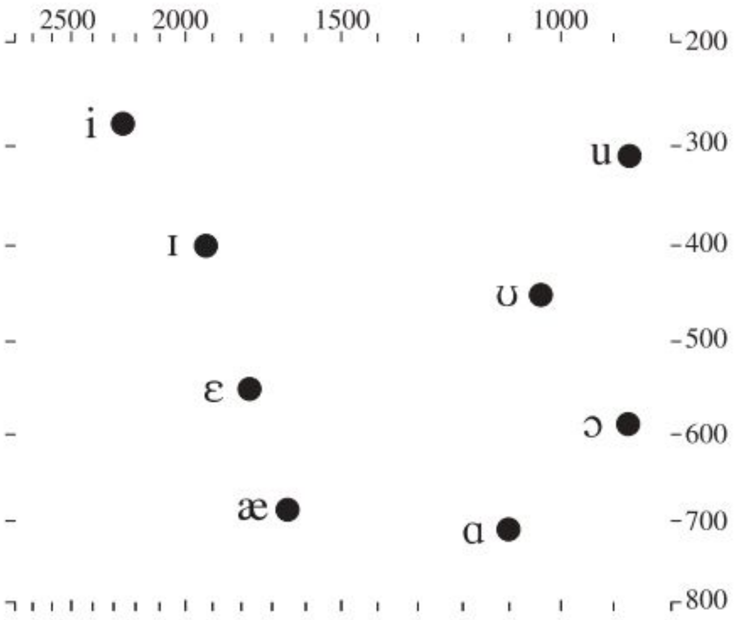
Figure 8.4 A spectrogram of the words *heed, hid, head, had, hod, hawed, hood, who'd* as spoken by a male speaker of American English. The locations of the first three formants are shown by arrows.



Ladefoged & Johnson (2006)

Sound as a frequency spectrum

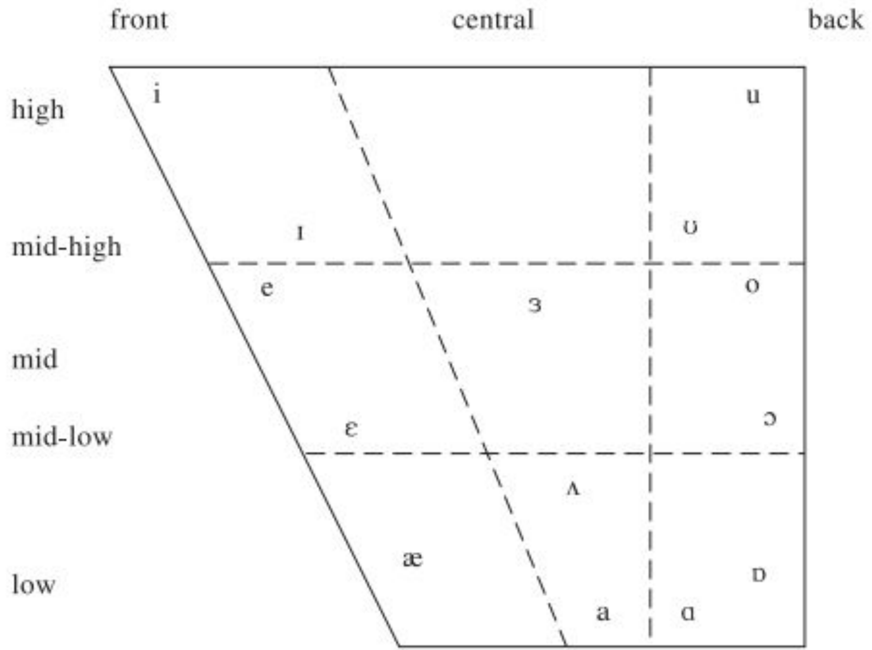
Figure 8.6 A formant chart showing the frequency of the first formant on the ordinate (the vertical axis) plotted against the second formant on the abscissa (the horizontal axis) for eight American English vowels. The scales are marked in Hz, arranged at Bark scale intervals.



Ladefoged & Johnson (2006)

Sound as a frequency spectrum

Figure 2.2 A vowel chart showing the relative vowel qualities represented by some of the symbols used in transcribing English. The symbols [e, a, o] occur as the first elements of diphthongs.



Ladefoged & Johnson (2006)

Sound as a frequency spectrum

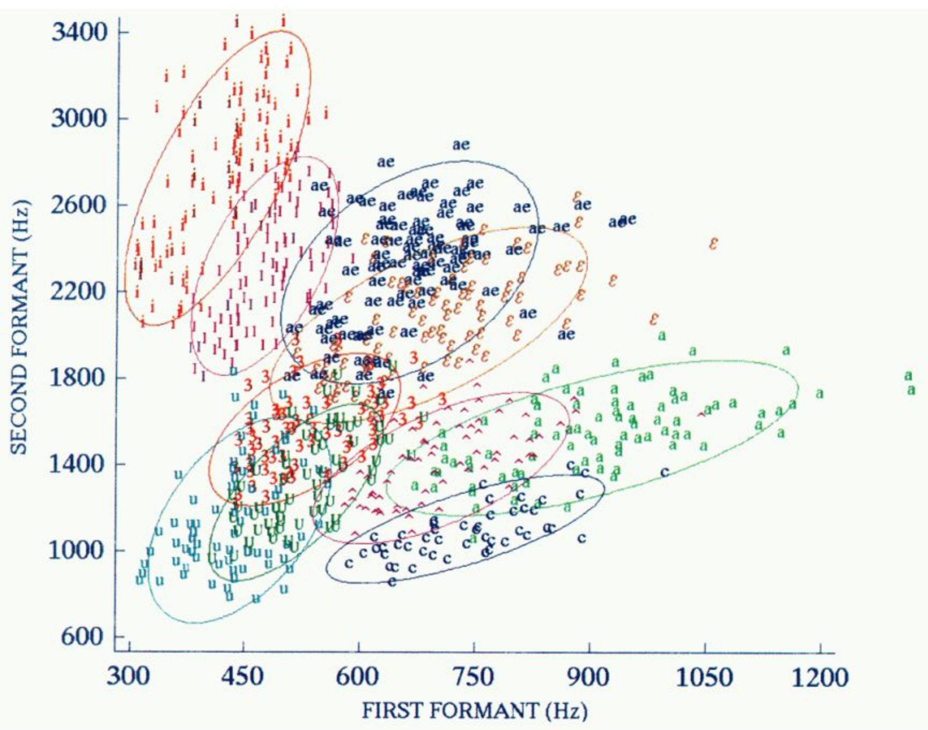
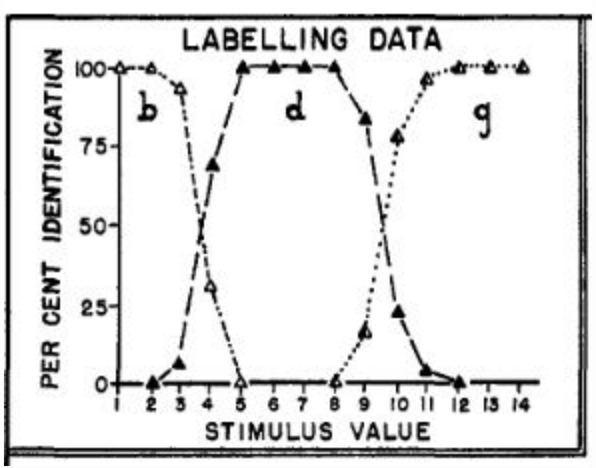


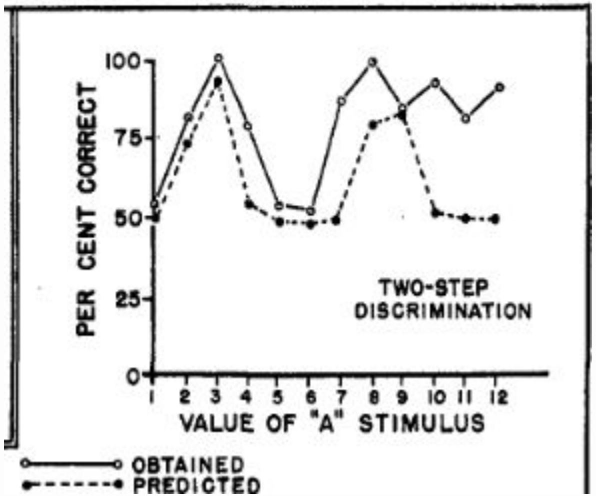
FIG. 4. Values of $F1$ and $F2$ for 46 men, 48 women, and 46 children for 10 vowels with ellipses fit to the data ("ae"=/æ/, "a"=/ɑ/, "c"=/ɔ/, "ʌ"=/ʌ/, "ɜ"=/ɝ/). Measurements for /e/ and /o/ have been omitted, and the data have been thinned of redundant data points.

Hillenbrand et al. (1995)

Categorical perception

Liberman et al. (1957): an identification and a discrimination task





Categorical perception

Main patterns seen as evidence of categorical perception

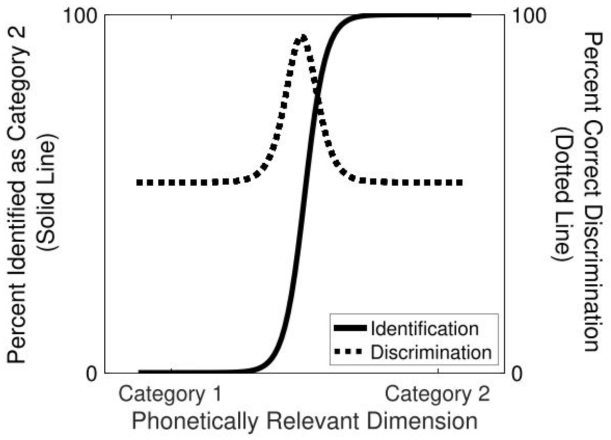


Figure 1. Hypothetical identification and discrimination functions in two-alternative forced choice tasks.

Figure from Feldman et al. (2021)

Representation learning

- Classic categorical perception is mostly a boundary effect
- But other related theories also predict structure within categories
- Both types describe category-driven *warping of perceptual space*

Perceptual magnet

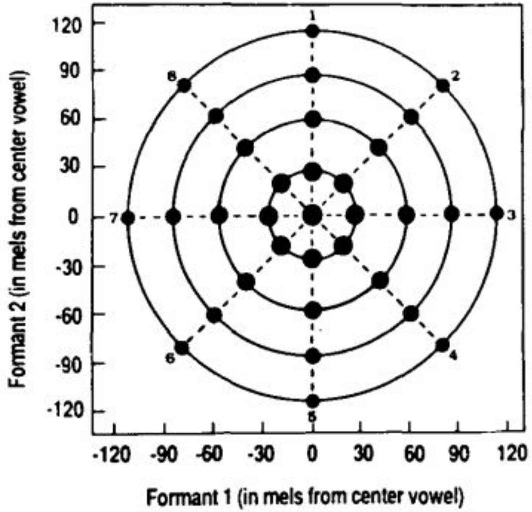


Figure 1. Formant frequency values in mels for stimuli surrounding a center vowel stimulus. The stimuli form four orbits and eight vectors around the center stimulus. The stimuli on each orbit are a specified distance in mels from the center vowel (30, 60, 90, or 120 mels, starting from the first orbit); the eight stimuli on each orbit differ in the direction and amount of formant frequency change.

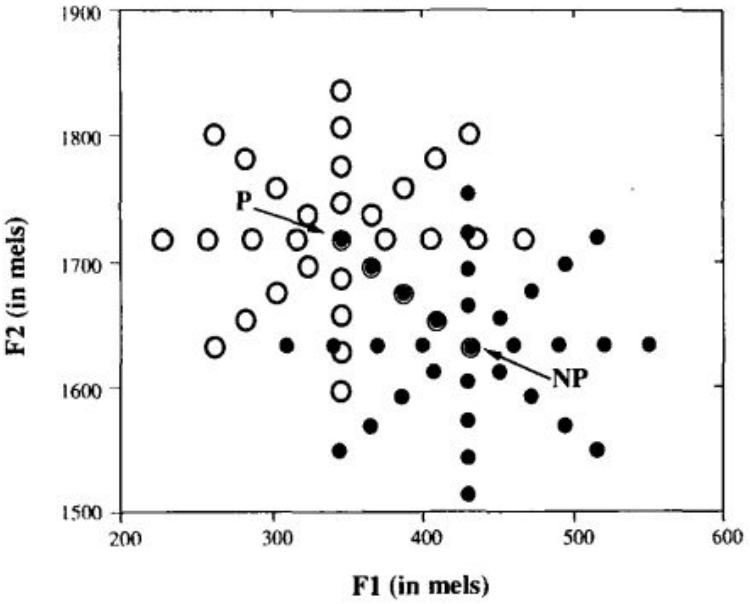
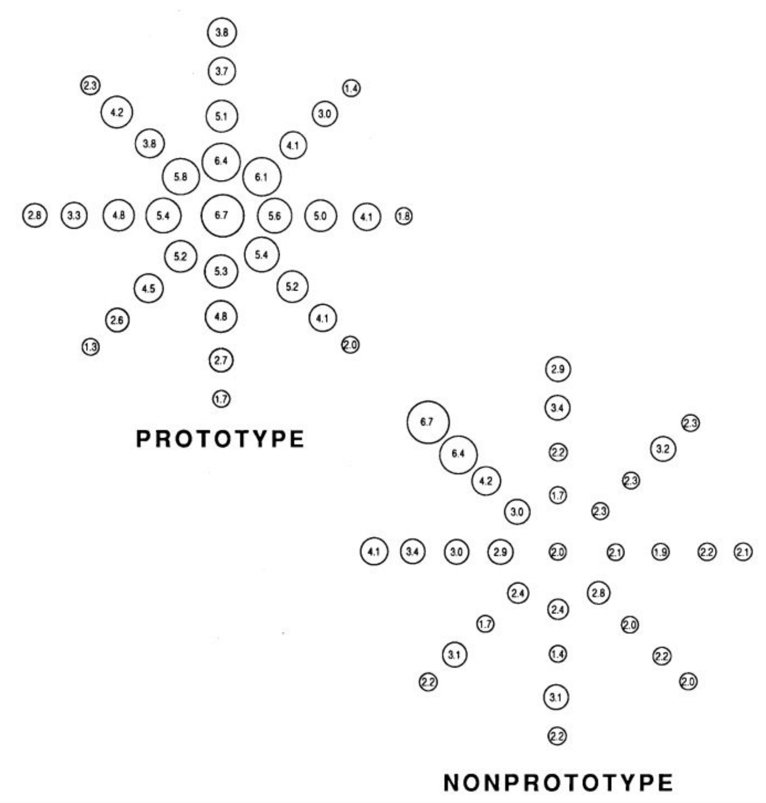


Figure 2. The prototype /i/ vowel (P) and variants on four orbits surrounding it (open circles) and the nonprototype /i/ vowel (NP) and variants on four orbits surrounding it (closed circles). The stimuli on one vector were common to both sets.

Kuhl (1991)

Perceptual magnet



Kuhl (1991)

Perceptual magnet

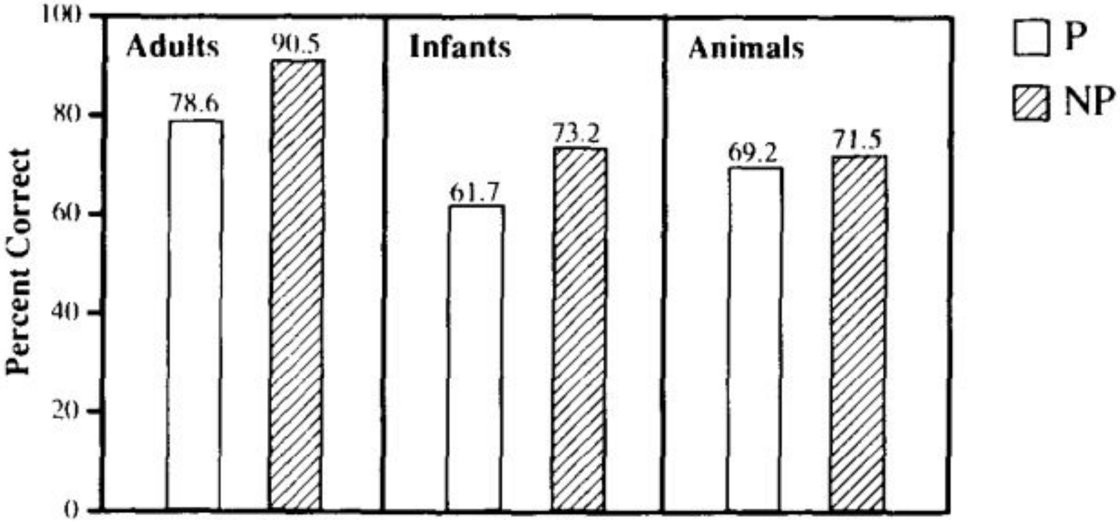
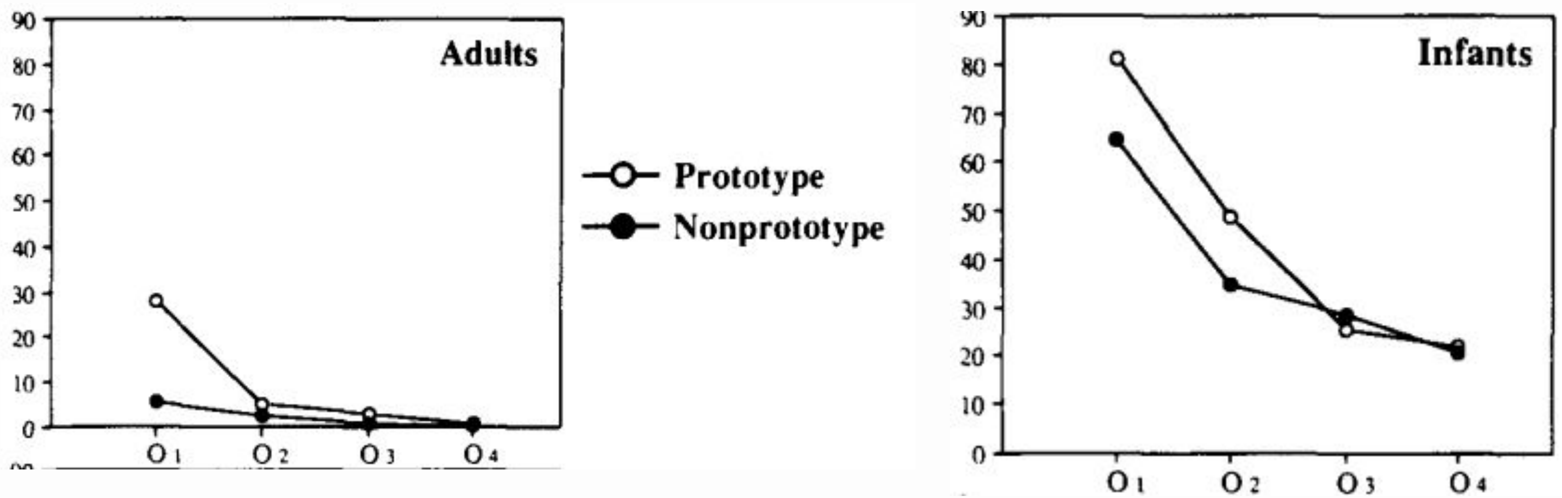


Figure 4. Average overall percent-correct scores achieved by adults (Experiment 2), infants (Experiment 3), and monkeys (Experiment 4) in the prototype (P) and the nonprototype (NP) conditions. For adults and infants (but not monkeys), there is a statistically significant difference between scores in the two conditions, with overall percent-correct scores being higher in the nonprototype condition.

Kuhl (1991)

Perceptual magnet



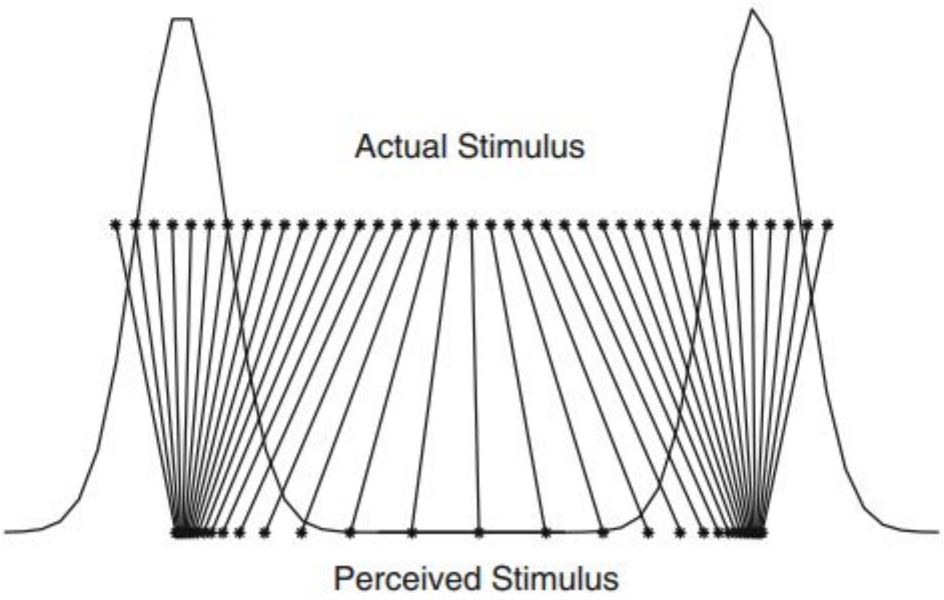
(Over)generalisation score

Kuhl (1991)

Perceptual magnet

Perceptual Magnet Effect (Kuhl, 1991)

prototypes pull nearby stimuli toward them, reducing discriminability near category centers



Kronrod et al. (2015)

Categories in early phonetic learning?

Early phonetic learning: Categories

The standard account: **phonetic category learning**

- Infants track the distribution of acoustic cues
- If the distribution is multimodal, categories are formed with centers at those modes, which then help restructure perceptual space

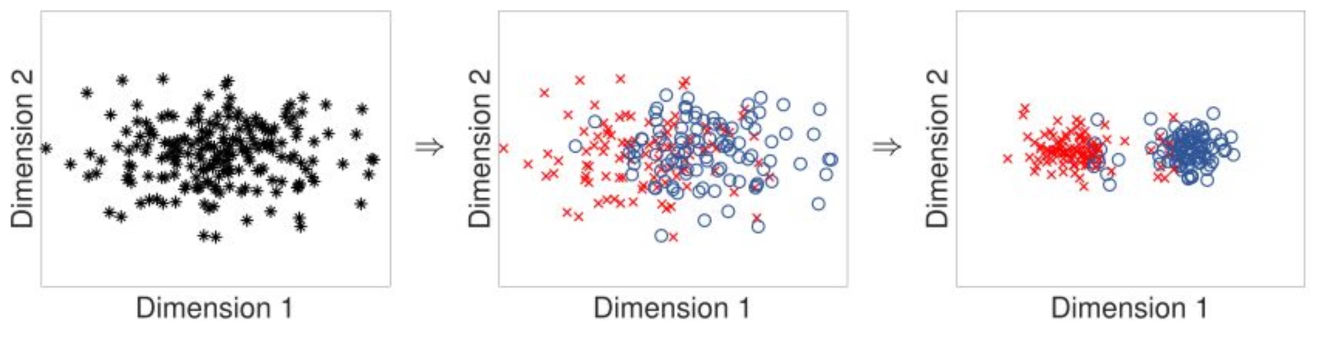
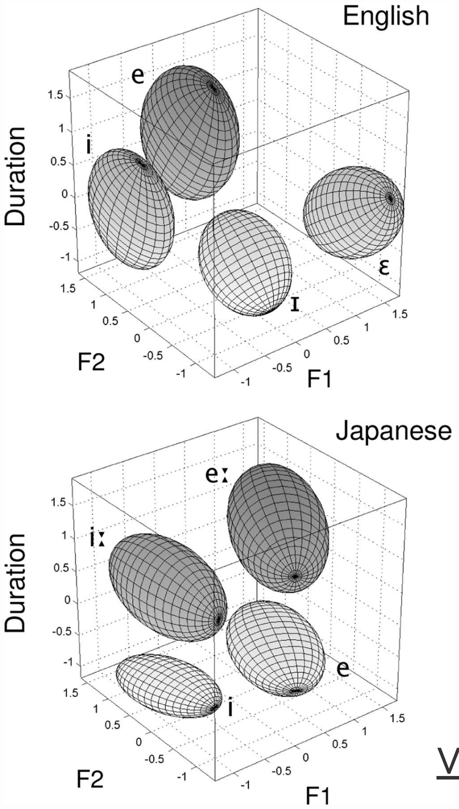
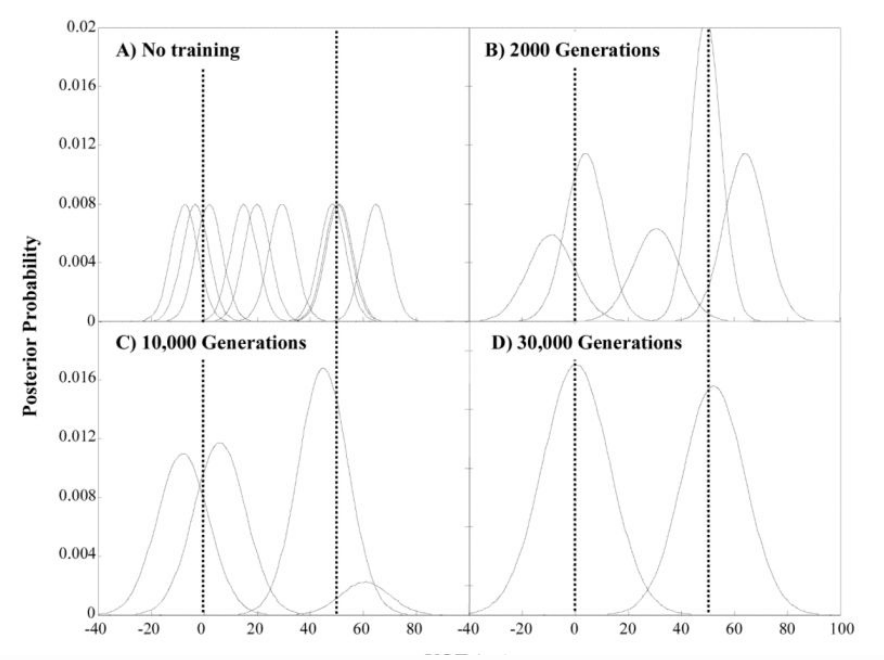


Figure from *Feldman et al. (2021)*

Early phonetic learning: Categories



Vallabha et al. (2007)

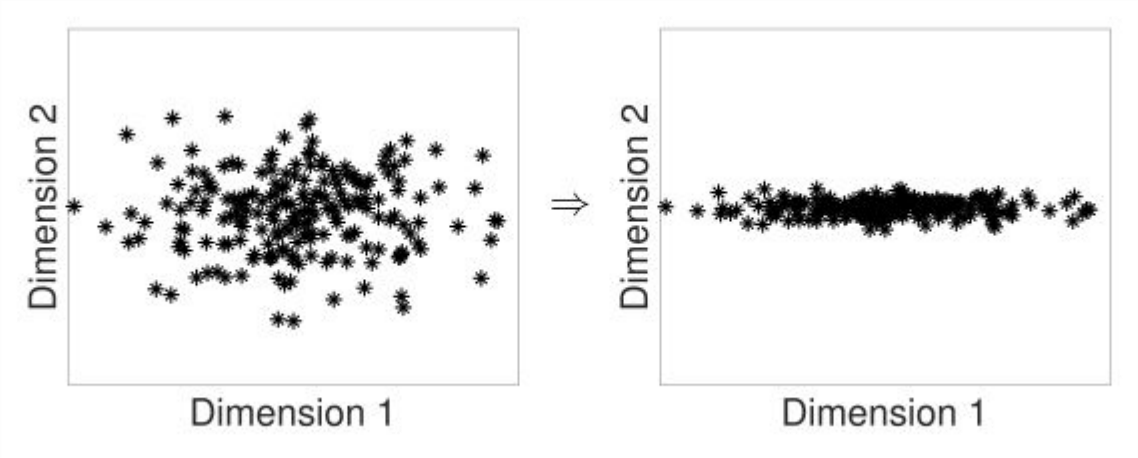


McMurray et al. (2009)

Early phonetic learning: Perceptual space

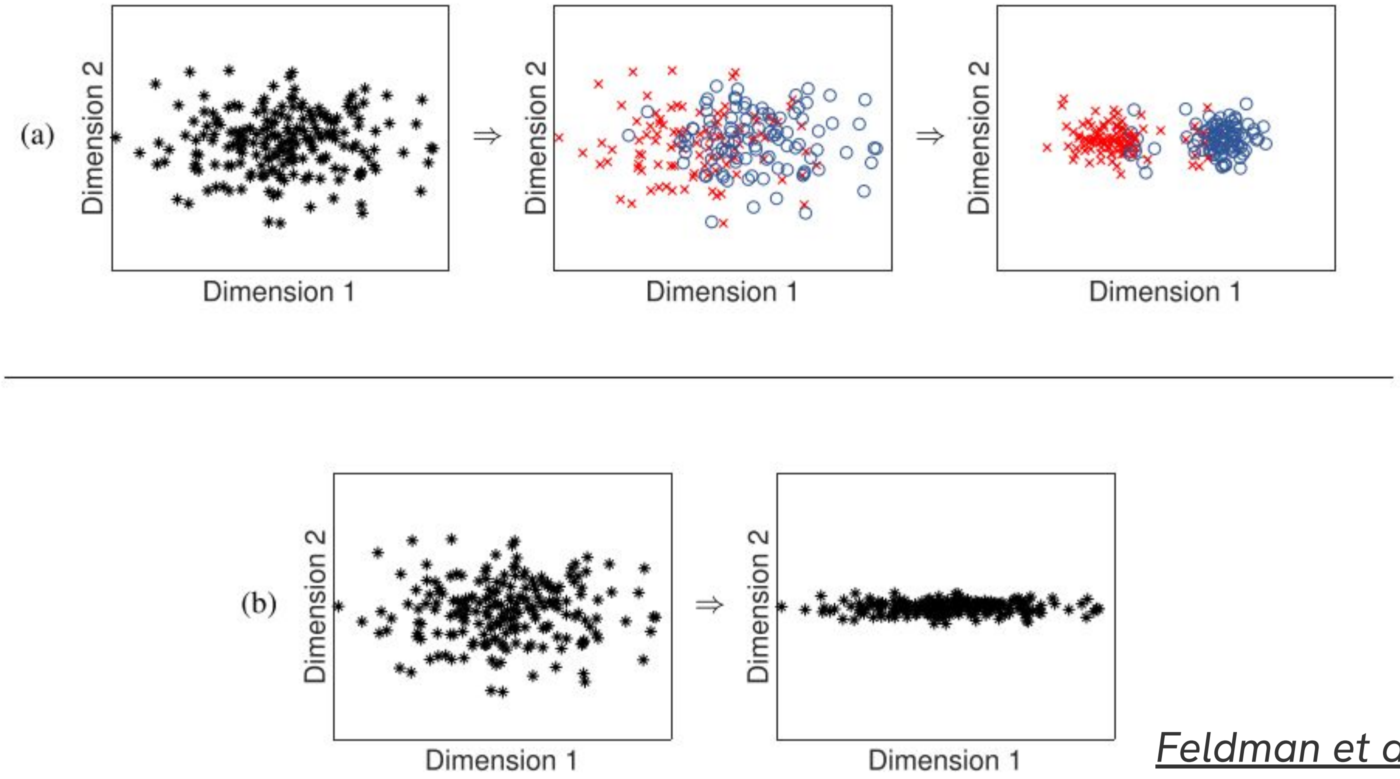
An alternative account: **perceptual space learning**

Children's perceptual space changes a lot before categories are formed



Feldman et al. (2021)

Early phonetic learning: Perceptual space



Feldman et al. (2021)

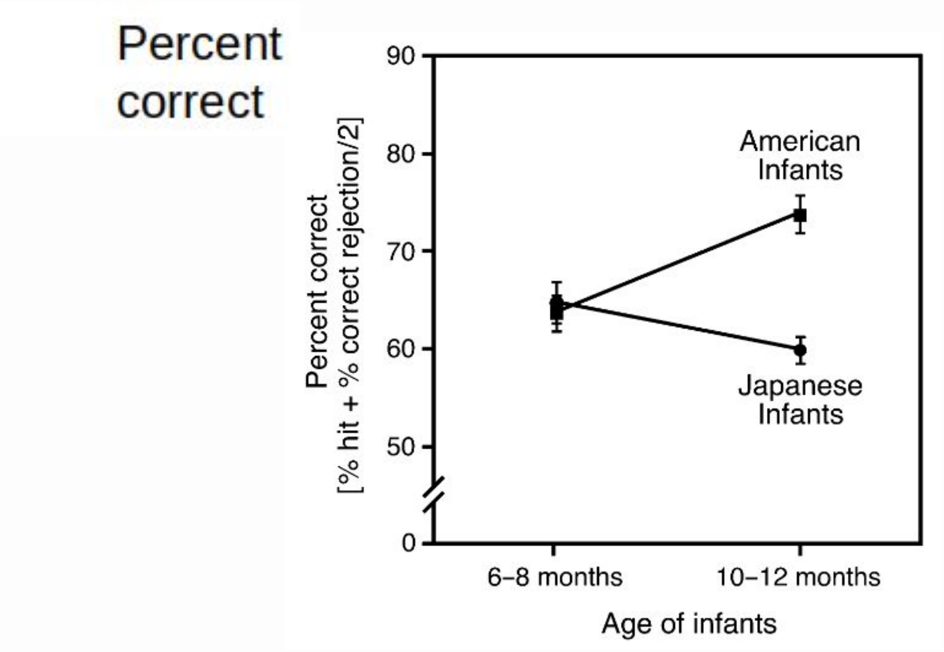
Testing the perceptual space account

English: [ɹ] ≠ [l]

- rock – lock
- wrong – long

Japanese: [ɹ] ≈ [l]

- no meaningful difference



Kuhl et al. (2006)

Testing the perceptual space account

- Similar data for sound contrasts in several language pairs (e.g., Bosch & Sebastián-Gallés, 2003; Tsao et al., 2004):
  - Mandarin (vs. English) infants: ɕ – $tɕ^h$
  - Catalan (vs. Spanish) infants: e – ɛ

Testing the perceptual space account

- How are infants commonly tested?
- Conditioned head-turn:
  1. Conditioning:
     Infant hears sounds and is taught to turn her head when the sound changes.
  2. Testing:
     Infant hears sounds, and we test whether she turns her head at the target contrast (e.g., [ɹ] and [l] for Japanese infants).

Testing the perceptual space account

Test: English [ɹ] – [l]

English

Japanese

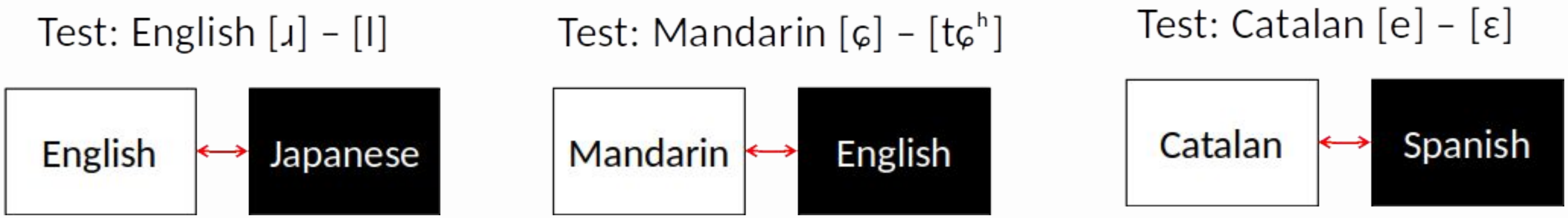
Training

[ɹ] – [l] discrimination

Native error *vs.* Non-native error

Expectation: native learner shows lower error than non-native learner.

Testing the perceptual space account



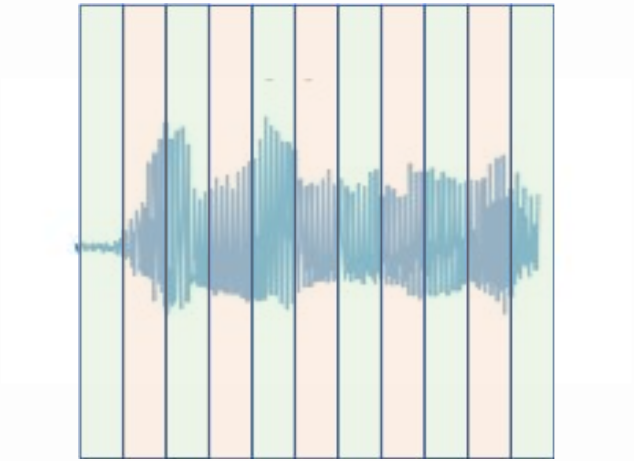
Same high-level setup to test 5 different model architectures

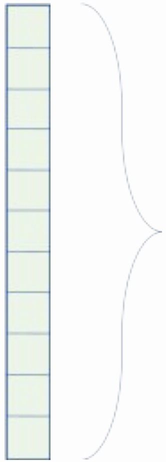
Testing the perceptual space account

Model	Weak supervision	Representation type
Dirichlet Process GMM [Chen et al., 2015]	–	25-ms-long frames
Autoencoder [Kramer, 1991]	–	25-ms-long frames
Correspondence autoencoder [Kamper et al., 2015]	+	25-ms-long frames
Autoencoder RNN [Chung et al., 2016]	–	Word-sized units
Correspondence autoencoder RNN [Kamper, 2019]	+	Word-sized units

Speech features

Speech features





39 numeric features: MFCC(+Δ+ΔΔ), standard in speech recognition

25 ms

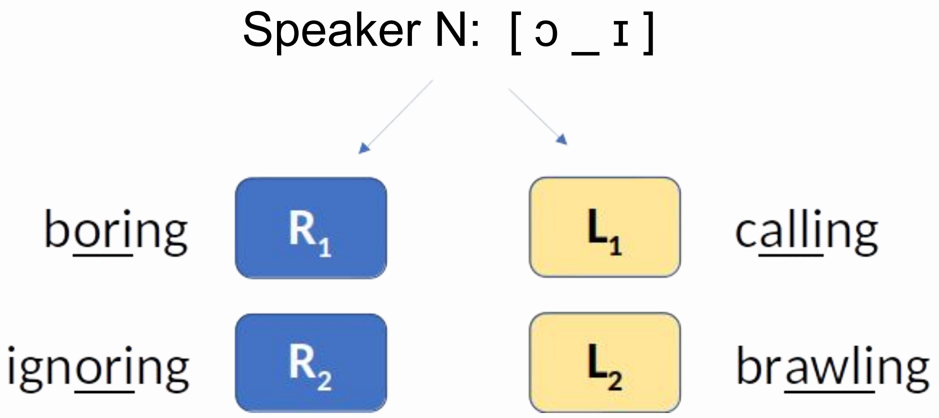
Testing the perceptual space account: Data

- Transcribed corpora of natural speech
  - WSJ, Buckeye, GlobalPhone, Glissando, AIShell.
- 2 different training samples per language.
- 7–58 hours of training data per sample.
- Native/non-native samples are matched on duration, number of speakers, gender.

Testing the perceptual space account: ABX

Machine ABX task (Schatz, 2016)

Extract target phones [ɹ] and [l] (from the same speaker in the same context)



Testing the perceptual space account: ABX

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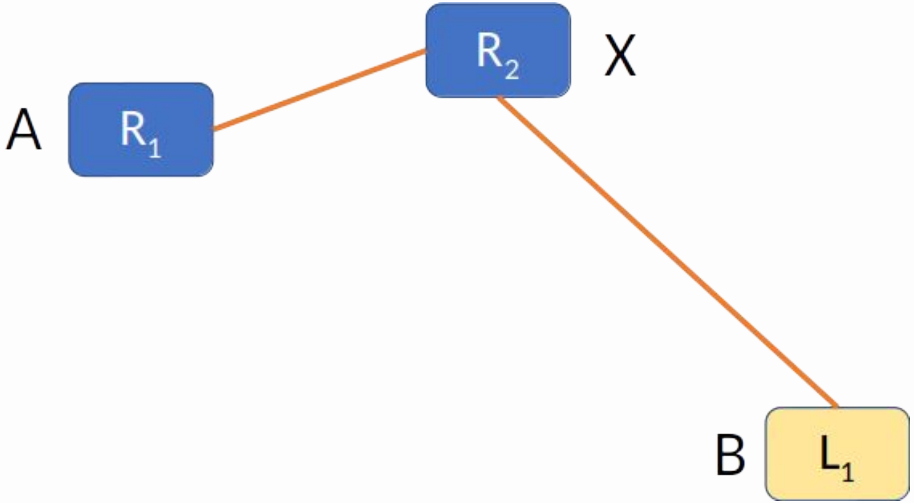
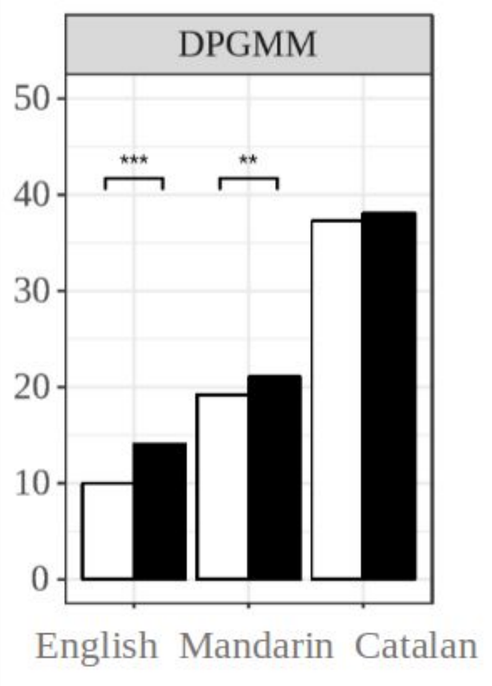


Testing the perceptual space account: ABX

ABX discrimination error



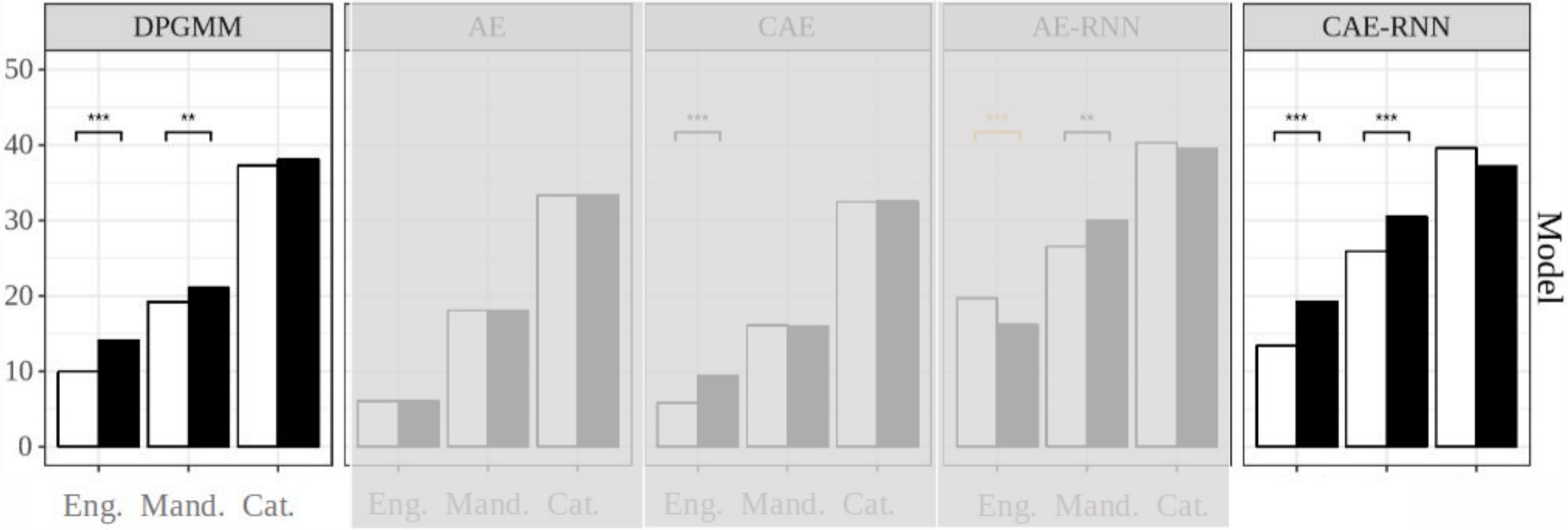
Test contrast

Native

Non-native

Matusevych et al. (2023)

Testing the perceptual space account: ABX



Matusevych et al. (2023)

Perceptual space account

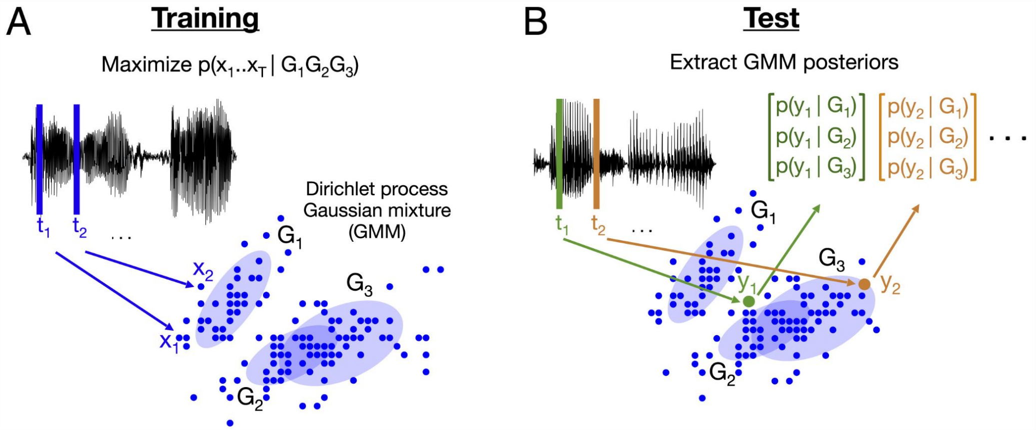
Feldman et al. (2021):

- Signature categorical perception behavior hasn’t convincingly been demonstrated in young infants
- Our models suggest that the phonetic discrimination effects can be explained without phonetic categories
- What is thought to be categorical behavior doesn’t necessarily imply discrete categories here

Model overview: DPGMM

Dirichlet process Gaussian mixture model (Schatz et al., 2021)

- Fully unsupervised bottom-up model
- Learns by clustering
- Representations at the level of individual speech frames

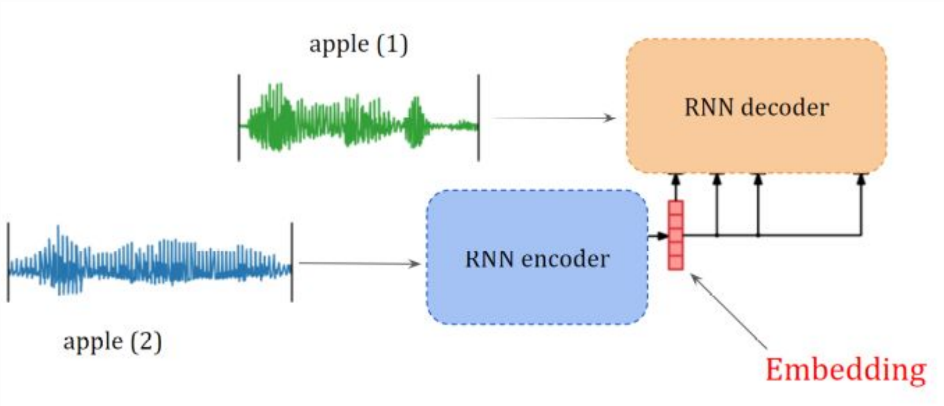


Schatz et al. (2021)

Model overview: CAE-RNN

Correspondence-autoencoding recurrent neural network (Kamper, 2019)

- Weak top-down supervision
- Learns by sequentially reconstructing an acoustic word
- Represents speech chunks of variable duration holistically



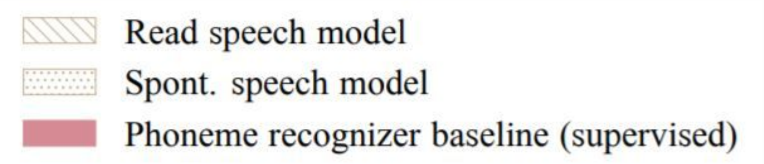
Kamper (2019)

Are there categories in the models?

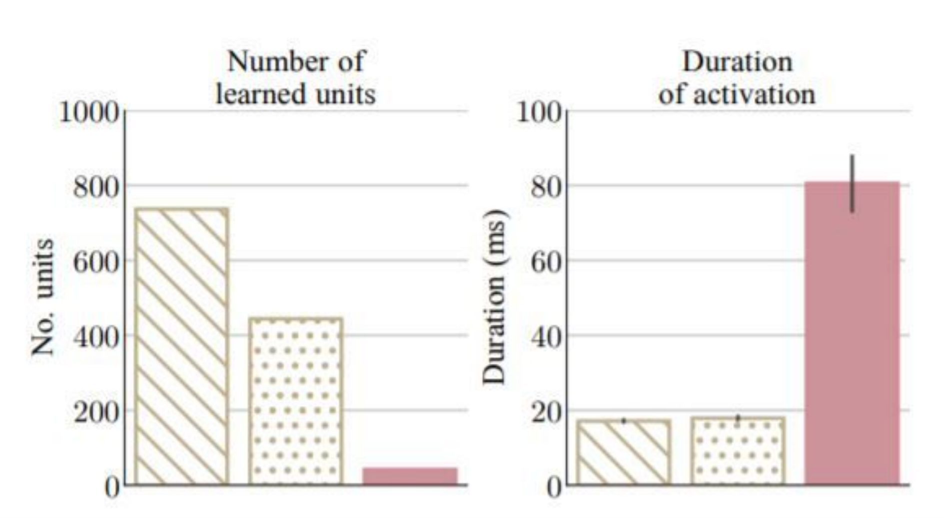
- Neither model is trained to learn categories.
- Can we still detect them in models’ representation spaces?

Are there categories in the models?

Schatz et al.'s (2021) DPGMM: Gaussians are not phonetic categories



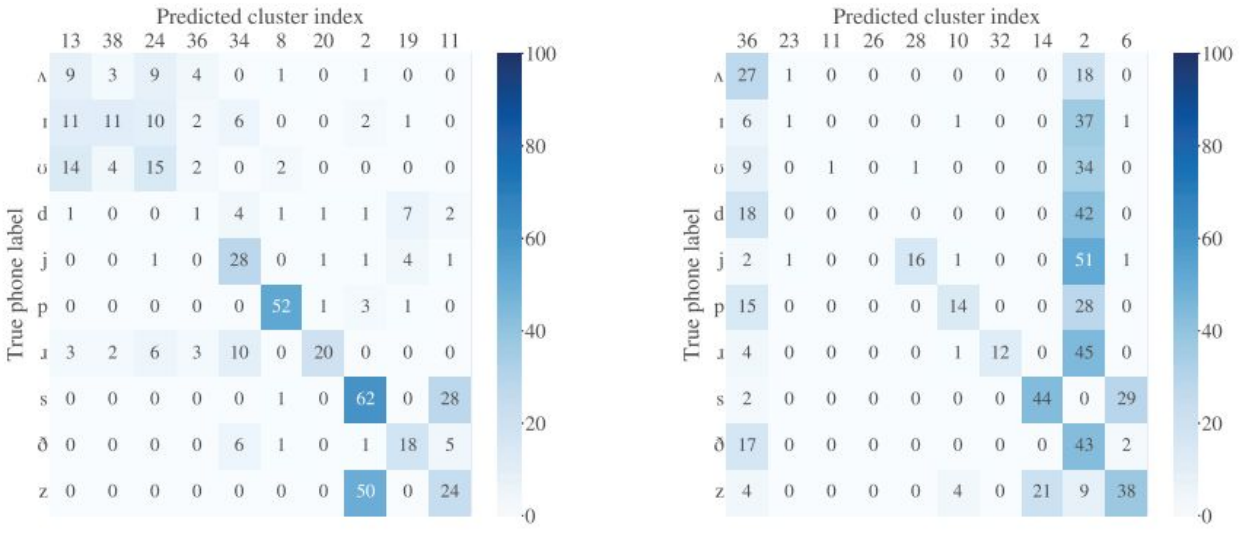
Schatz et al. (2021)



Are there categories in the models?

- Run agglomerative clustering on individual phones encoded into model's representation space
    - Posteriorgrams for DPGMM
    - Acoustic embeddings for CAE-RNN
- Find the best mapping between clusters and true labels

Are there categories in the models?



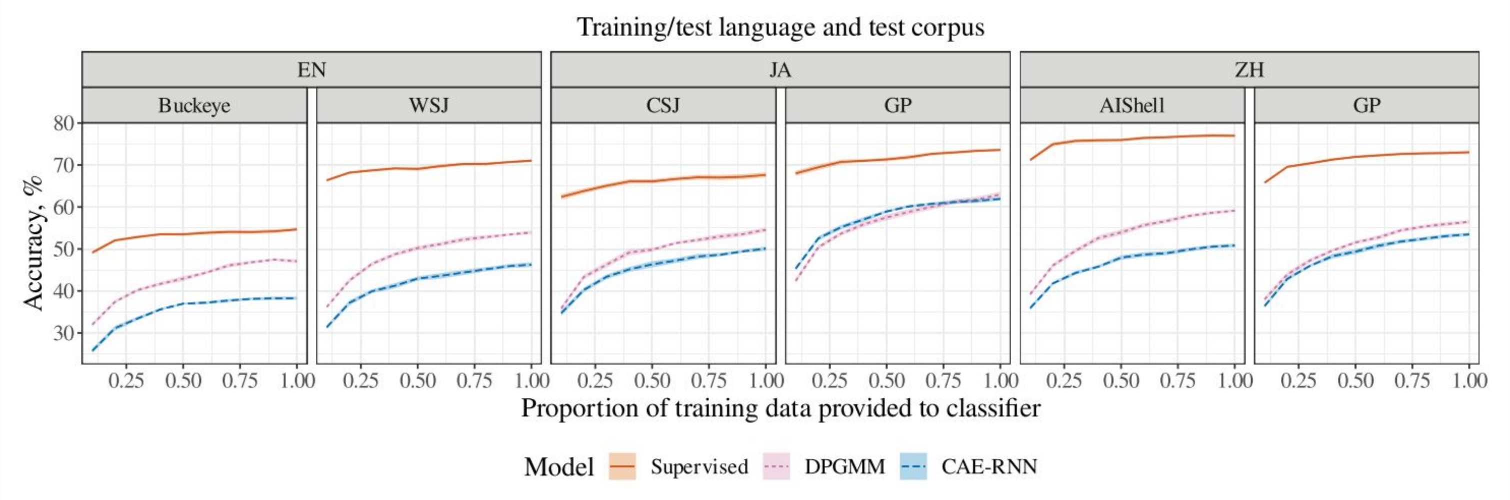
(a) DPGMM

(b) CAE-RNN

Matusevych et al. (2023)

Are there categories in the models?

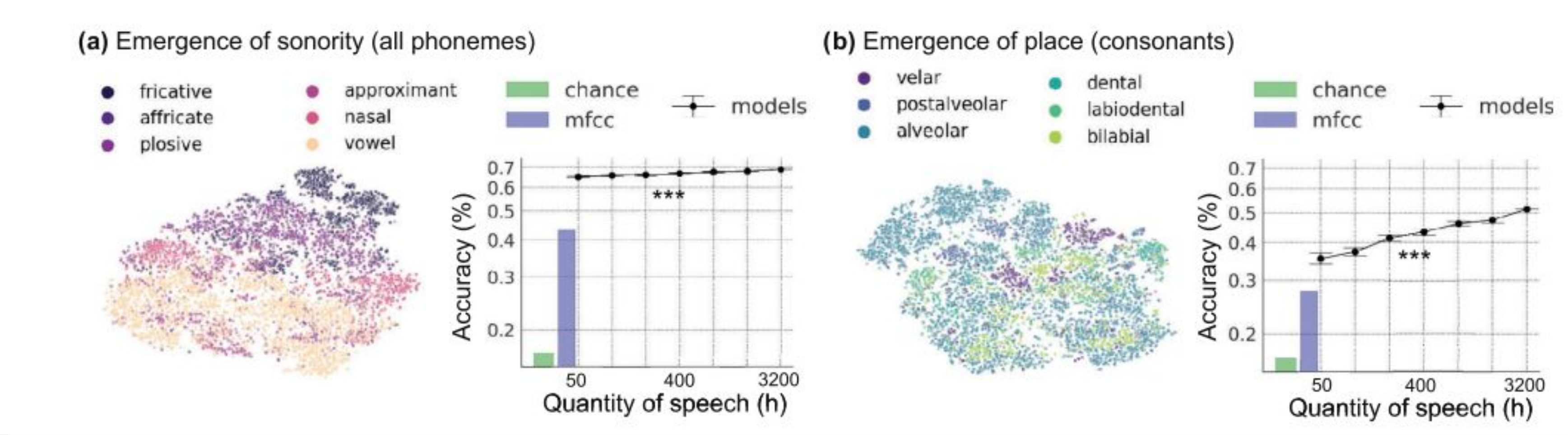
- Train a KNN classifier on model representations to categorize phones



<u>*Matusevych et al. (2023)*</u>

Are there categories in the models?

- Lavechin et al. (2024) - a model of phonetic and word learning



Lavechin et al. (2024)

Are there categories in the models?

- Lavechin et al. (2024) - a model of phonetic and word learning

  "as the system is provided with more data, the learned representations exhibit a progression towards greater linguistic structure <...> However, this is still far from showing the emergence of actual linguistic categories."

Are there categories in the models?

What does it mean for a model to represent a category?

From weaker to stronger evidence:

A. The information is recoverable: a probe can decode it
B. The information is in a usable format: e.g. linearly separable, cheap to read out
C. The model actually uses it to drive behavior

But: What is that “information” we are looking for?

Are there categories in the models?

- **General consensus:** linguistic structure emerges in neural network models
- **Speculation:** the concept of linguistic categories might be underspecified, making it difficult to be looking for them in model representations

Are there categories in the models?

- What could count as strong evidence of categories?
  - A model with categories explicitly encoded (by design) predicts the data correctly, but a model without them does not
  - A sharp identification function together with a discrimination peak at the boundary **and** poor within-category discrimination, with discrimination predictable from identification

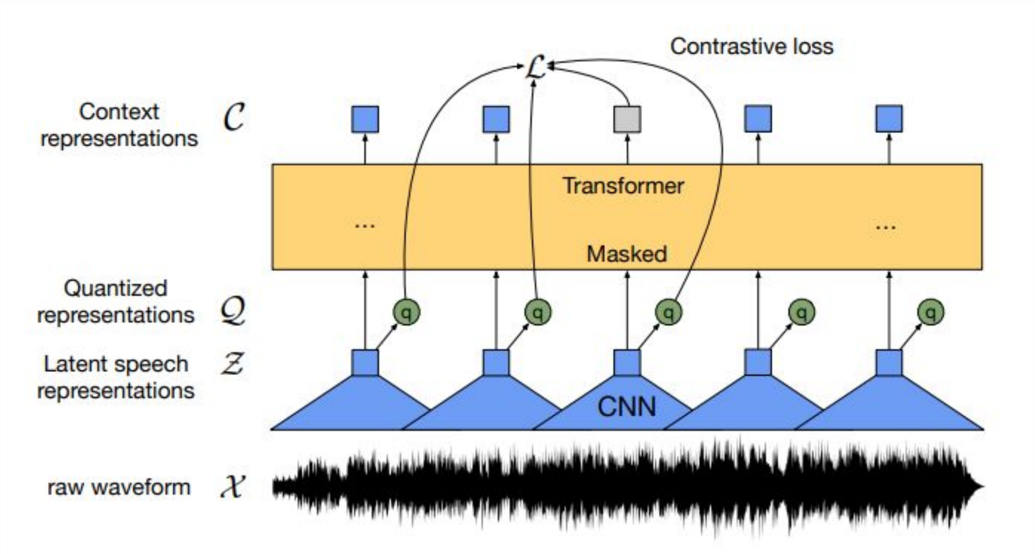
Phone discrimination in a large speech model

Phone discrimination in a speech model

- Probe a large speech model
- Questions:
  - Where in the model is phonetic information is represented?
  - Can we detect categorical perception in the model?

Phone discrimination in a speech model

- Model: wav2vec 2.0, trained to predict masked segments of speech from raw audio.



Baevski et al. (2020)

Phone discrimination in a speech model

- Model: wav2vec 2.0, trained to predict masked segments of speech from raw audio.
- Self-supervised: learns from audio alone, with no phonetic labels and no category structure built in.

Baevski et al. (2020)

Phone discrimination in a speech model

- Data: Hillenbrand et al. (1995):
  - Recordings of 12 American English vowels, each produced in an h-V-d word (*heed, hid, head*, and so on)
  - 45 men, 48 women (plus 46 children, not used here)

<u>*Baevski et al. (2020)*</u>

Phone discrimination in a speech model

- Approach:
    - Extract internal representation at every layer: the input projection plus 12 transformer layers
    - Measure how well each layer tells vowels apart: for each vowel token, average the vectors over time, giving one vector per token per layer.

Phone discrimination in a speech model

- ABX task as explained earlier:
    - Take three tokens: A and X are the same vowel, B is a different vowel.
    - The representation is correct on this triplet if X is closer to A than to B, using distance in the representation space.
    - Average over many triplets to get a score per layer. Chance is 0.5; perfect discrimination is 1.0.

Phone discrimination in a speech model

- Testing categorical perception:
  - A continuum of synthesized stimuli from *lih* to *rih* (de Heer Kloots & Zuidema, 2024)
  - Testing identification in our model (should be sigmoid: consistent within a category, sharp switch at the boundary)
  - Testing discrimination in our model (should be low and flat within categories, peaking at the boundary; and should follow the pattern predicted by identification)

Thanks!